

Chapter 41 — Management Strategies for Vesicoureteral Reflux

- **Vesicoureteral reflux (VUR)** is the **retrograde flow of urine from the bladder to the upper urinary tract**. It is associated with **acquired scarring** and with **congenital renal dysplasia**.
- **Historically managed dichotomously; now viewed as a spectrum of severity**, which calls for **risk-stratified management**. Despite several large randomised trials, aetiology, diagnosis and treatment all remain contested.

EMBRYOLOGY OF THE URETEROVESICAL JUNCTION

- **Weeks 4–6: the ureteric bud arises from the mesonephric (wolffian) duct**. The early ureter and vas form the two arms of a Y, with the common mesonephric duct as the stem.
- The stem is absorbed into the **urogenital sinus (UGS)** until the two arms enter separately: the **ureter into the bladder**, the **vas and ejaculatory duct into the prostatic urethra** (in the female, the vestigial **Gartner duct** in the vagina).
- **Mackie and Stephens**: the arms **rotate** on contact with the UGS, so the **ureteric orifice ends up proximal to the ejaculatory duct orifice**.

	Early budding	Late budding
Contact with UGS	Too soon → over-rotation	Too late → insufficient rotation
Orifice position	High and lateral	Distal and medial
Result	Short intramural tunnel → VUR	Ectopic ureter , often obstructed at the bladder neck or beyond
Kidney	Abnormal bud–metanephros contact → dysplasia, hypoplasia or agenesis	

- The nephric duct may **not form the trigone but undergo apoptosis**, a pivotal step in ureteric transposition, possibly driven by **vitamin A-induced signals** from the primitive bladder.

Concept — One event, bud position, sets both the orifice and the kidney. That is why high-grade reflux and renal dysplasia travel together and why some “scars” are congenital.

FUNCTIONAL ANATOMY OF THE ANTIREFLUX MECHANISM

- VUR reflects an **imbalance among three factors: (1) the ureter**, a dynamic conduit whose **peristalsis actively opposes reflux**; (2) the **anatomy of the ureterovesical junction (UVJ)**; (3) **bladder dynamics**.
- **The intramural ureter** runs within the detrusor. At the hiatus, the **outer ureteric muscle merges with outer detrusor to form Waldeyer's sheath**, contributing to the deep trigone. Recent immunohistochemistry suggests Waldeyer's sheath may be a feature of the **whole ureter**, not unique to the UVJ.
- **Passive flap-valve**: the intramural ureter is **compressed by the bladder wall during filling**. This needs **adequate intramural length and fixation**.
- **Paquin**: **tunnel length to ureteric diameter about 5:1 in non-refluxing UVJs versus 1.4:1 in refluxing ones**. Later work and computer modelling add that **orifice configuration** matters as much as length. Both theories neglect the **intrinsic qualities of the ureter**, its ability to collapse with filling.
- **Opening (active)**: the longitudinal tunnel muscle contracts → the tunnel **shortens and widens** → the bolus passes. Cystoscopically, the **orifice moves laterally with each jet**. **Closure**: compression plus return to full tunnel length as the muscle relaxes.

Figure legend, Fig. 41.1 — A refluxing UVJ has the same anatomy as a non-refluxing one except for an inadequate intravesical submucosal length. Marginal tunnels may reflux intermittently. Superficial and deep periureteral sheaths continue beyond the orifice into the trigone, and no true space separates ureter from bladder (Glenn, 1975; Elbadawi, 1972).

KEY POINTS: FUNCTIONAL ANATOMY

- Antireflux integrity depends on the **balanced anatomical and functional relationship between ureter and bladder**.
- VUR may be **anatomical or functional** in origin, at the UVJ, bladder or outlet.

Concept — The UVJ is a **dynamic valve, opened actively and closed passively**. Reflux follows whenever tunnel geometry, ureteric function or bladder pressure falls out of balance.

DEMOGRAPHICS

Prevalence

- Prevalence is hard to measure because VUR is **sometimes transient**. Historical screening found VUR in about a **third of hospitalised children without urological disease**, higher in infants. Screened normal newborns showed **1–4%** (interpret with caution). Up to **38%** occurs with postnatal upper-tract sonographic anomalies.
- **Best estimate (meta-analysis): about 30% of children with urinary tract infection (UTI), and about 17% of those imaged for other reasons** such as hydronephrosis.

Disparities Related to Sex, Age, and Race

DEMOGRAPHIC MODIFIERS OF VUR

Factor	Finding	Explanation
Sex	Overall, girls about twice as often . But in infancy males predominate (up to 3:1 at 0–6 months) , shifting to 1:1 by about 2 years ; most males present <1 year	Uncircumcised infant boys have 10–12-fold more UTI , so they are investigated more. Infant male voiding pressures are higher (see Aetiology). A non-retractable foreskin raises UTI risk over eightfold in boys with VUR
Age	Younger = more VUR ; resolves with time	UVJ maturation
Race	Less common in Black children (historically up to tenfold, now about threefold); no grade IV–V in Black children under 6 ; VUR absent in Black infants followed for antenatal hydronephrosis in one series; equal after age 10	Faster UVJ maturation; socioeconomic and ancestry factors; genetic ancestry data are now blurring “race”

- **International Reflux Study in Children (IRSC): only 10% of US entrants were male versus 24% of European**, reflecting **circumcision rates of 62% (US) versus 5% (Europe)**. This supports UTI-driven detection bias.

TABLE 41.2 INCIDENCE OF REFLUX IN PATIENTS WITH URINARY TRACT INFECTIONS

Age (years)	<2	2–6	7–11	12–21
Incidence of VUR with UTI (%)	38.6	26.6 (text: 26.9)	19.7	7.6

From Chand DH, Rhoades T, Poe SA, et al., J Urol 2003.

- **What it tests:** the **steady fall of VUR with age among children with UTI**, from about 4 in 10 under 2 to under 1 in 10 in adolescence. The trap: the same UTI in an older child is **far less likely to reveal reflux**. The 2–6-year value differs between table and text because of a **transcription difference in the source**.

KEY POINTS: DEMOGRAPHICS

- **The younger the child with UTI, the likelier VUR.**
- **Males have more UTI early, but VUR is diagnosed more often in girls presenting with UTI.**
- **VUR is relatively rare in Black children**, though genetic-ancestry data are blurring historical racial categories.

Vesicoureteral Reflux in the Fetus

- **Prenatal hydronephrosis occurs in 1–5% of pregnancies.** Routine VCUG screening of all such infants has been **abandoned**.
- **Voiding cystourethrogram (VCUG) in the newborn only for higher-risk renal bladder ultrasonography (RBUS) findings:** higher-grade (especially bilateral) hydronephrosis, cortical abnormality, ureteric dilatation, suspected ureterocele or duplication, bladder abnormality.
- VUR is found in about a **fifth of those with persistent postnatal hydronephrosis** who reach VCUG, mostly with higher grades. Older screening series found **10–40%**, with **bilateral dilating VUR far commoner in high-grade prenatal hydronephrosis**.
- **Clinical significance is usually low:** few infants with screen-detected VUR go on to UTI. Infants not screened who later have a febrile UTI frequently prove to have VUR. The implication is to **individualise** rather than VCUG every hydronephrotic infant.

INHERITANCE AND GENETICS

Sibling Vesicoureteral Reflux

- **Sibling prevalence about 30% (up to 50%),** higher in **younger siblings**. Almost 90% of screened siblings **under 3** had VUR in one series. Symptomatic siblings had more high-grade VUR and scarring. In **older siblings**, prevalence may fall to **7%**, consistent with spontaneous resolution, though not all series agree.
- That series recommended **screening siblings under 3 of index patients with grade III–IV**, but it cannot guide siblings of grade I–II index cases, who are the majority.
- **Identical twins:** very high, even **complete, concordance**. Screen-detected twins had higher grades but **no scarring**, unlike their index twins.
- Screened refluxing siblings showed **renal abnormalities**, more often over age 2, which was read as progressive damage. But this **ignores early congenital damage magnified by growth** and **does not separate dysplasia from acquired scar**.
- **The real target is renal damage, not VUR itself.** Screen non-invasively for **cortical abnormality**, and reserve VCUG for **UTI or bladder and bowel dysfunction (BBD)**. An **older sibling with normal kidneys and no BBD** gains little from detection: treat a febrile UTI as in any child.
- **American Urological Association (AUA) guideline:** VCUG for siblings with **cortical abnormality or renal size asymmetry on ultrasound, or a UTI history**. This is especially so with **grade 3–4 hydronephrosis**; observe grade 1–2.

KEY POINTS: SCREENING, INHERITANCE, AND GENETICS

- Sibling prevalence is raised; multiple genes are involved (detailed online).
- Cortical abnormalities in refluxing siblings are not all acquired.
- Mass sibling screening is unnecessary: be selective by risk profile.

Concept — Sibling screening hinges on **what is being looked for**. VUR alone is common, transient and mostly harmless; **renal damage and infection risk** are what matter, so screening starts with ultrasound, not the catheter.

GRADING OF VESICoureTERAL REFLUX

International Classification of Vesicoureteral Reflux

- Proposed by the International Reflux Study Committee (1981). It grades the VCUG appearance of **ureter, pelvis and calyces** into five grades.

TABLE 41.3 INTERNATIONAL CLASSIFICATION OF VESICoureTERAL REFLUX

Grade	Ureter	Pelvis and calyces
I	Into a non-dilated ureter only	Not reached
II	Not dilated	Reached, no dilatation
III	Mild-moderate dilatation	Mild-moderate dilatation, minimal forniceal blunting
IV	Moderate tortuosity	Moderate dilatation
V	Gross dilatation and tortuosity	Gross dilatation; papillary impressions lost

- **Pattern:** I–II are non-dilating (the ureter alone, then the pelvis); III–V are dilating, with progressive calyceal change. **The step from II to III is the first dilatation; IV adds tortuosity; V loses the papillary impressions.**
- **Shortcomings:** ureteric and calyceal dilatation are **not always concordant** (Fig. 41.4). **Interobserver variability is large:** in RIVUR, all three readers agreed in only about 60% of ureters, worst for grades II–IV and best at the extremes.
- **Radionuclide cystography (RNC)** cannot assign international grades; it collapses them into **three grades** (Fig. 41.5). **Contrast-enhanced voiding urosonography** has no adopted grading system yet.
- **Grade is a snapshot**, modulated by bladder dynamics, outlet, diuresis, filling volume, the number of cycles, filling versus voiding reflux, and ipsilateral UVJ obstruction. None of these is built into the grading.

Figure legends, Figs 41.4–41.5 — A ureter with marked lower-segment dilatation but an undistorted collecting system **does not behave like typical grade II**. On **nuclear cystography**: grade 1 = international I; grade 2 = II–III; grade 3 = IV–V.

Other Tools

Ureteral Diameter Ratio

- **Ureteral diameter ratio (UDR)** = largest ureteric diameter within the false pelvis ÷ distance from L1 to L3 vertebral bodies.
- It reduces grading variability, especially for **intermediate grades**. It **correlates with grade, predicts spontaneous resolution better than grade**, and **predicts breakthrough UTI (BUTI) independent of grade**. It has been built into resolution and BUTI models and may predict **endoscopic injection success**.

Machine-Learning Models

- Image-analysis algorithms trained on standard image repositories show **promise for objective grading**.

Concept — Grade is a **static picture of a dynamic process, read inconsistently**. Hence the push towards measured, normalised variables (UDR) and multivariable risk tools that predict behaviour rather than describe appearance.

ETIOLOGY OF VESICoureTERAL REFLUX

- VUR is an **imbalance between bladder storage and voiding dynamics and the architectural competence of the UVJ**. VUR is also called secondary if it was **previously documented as absent**.

	Primary VUR	Secondary VUR
Fault	Congenital deficiency of the UVJ , with a normal low-pressure bladder	A normal-ish UVJ overwhelmed by pressure, or anatomically abnormal
Contributors	Short tunnel, weak detrusor backing, dilated or stiff ureter , bladder characteristics	Mechanical outlet obstruction (posterior urethral valves (PUVs), stricture, mass); functional obstruction (neurogenic dyssynergia; non-neurogenic dysfunctional voiding); UVJ anomaly (ureterocele, duplication, paraureteral diverticulum , iatrogenic)
Implication	A dilated ureter traditionally needed a long tunnel (>5 cm) or tapering ; a strict 5:1 ratio may not be essential	Treat the cause ; a tailored approach alters the trajectory

Primary Vesicoureteral Reflux

- It is caused by **more than tunnel length**, as set out in the table.

Secondary Vesicoureteral Reflux

- **Spina bifida neurogenic bladder: VUR in up to 30%**. Look for **occult dysraphism** in a child with UTI or incontinence: **sacral dimple, hairy patch, gluteal cleft anomaly, reduced anal tone, leg deficit, refractory constipation or encopresis**.
- Known dysraphism plus VUR → **proactive management: early clean intermittent catheterisation (CIC), anticholinergic or β_3 -agonist**, serial imaging, and a **low threshold** to investigate any progression.
- **Infant (especially male) secondary VUR**: high voiding pressures from **inadequate sphincter relaxation**, with overactivity or obstructive voiding patterns, mostly in boys. Filling is largely normal. This **resolves as voiding matures**, which explains the **infant male predominance, high male resolution and the sex shift with age**.
- **Older children: BBD** → rising voiding pressures and incomplete emptying → UTI. The pressures **distort the UVJ** and can create secondary VUR. **Normal bladders void at about 100 cm H₂O without reflux**, so structural UVJ failure is the key. **Reflux at low pressure or early filling = poor prognosis** for resolution.
- **Detrusor overactivity is the commonest urodynamic abnormality** with VUR in neurologically normal children. Targeted treatment (**timed voiding, biofeedback, anticholinergics, α -blockers**) makes resolution or improvement by two or more grades more likely. **Anticholinergics can eliminate VUR** in a substantial share.

- **Primary and secondary VUR are not mutually exclusive:** “primary” reflux often has a **secondary, treatable component** (overactivity, BBD).

Concept — Every refluxing UVJ sits somewhere between **an intrinsically weak valve and a normal valve under excessive pressure**. Identifying the pressure component is what makes VUR modifiable.

DIAGNOSIS AND EVALUATION OF VESICoureTERAL REFLUX

Confirmation of Urinary Tract Infection

- **Acquired reflux nephropathy is largely preventable** and requires **infected urine plus VUR**. Major decisions, including surgery, often rest on the diagnosis of UTI, so **proper urine collection and confirmation are paramount**.

KEY POINTS: DIAGNOSIS AND EVALUATION OF VESICoureTERAL REFLUX

- **Accurate urine collection and UTI diagnosis** avoid false positives and management errors.
- **The gold-standard VUR study requires bladder catheterisation.**
- **Nuclear cystograms are more sensitive but give much less anatomical detail** than VCUG.
- **Parental perceptions** must be considered.
- **Upper-tract assessment is serial.**
- The imaging challenge is to **separate congenital dysplasia from post-infective scarring**.

Evaluating Urinary Tract Infection

- VUR is found in **about 30% of children after UTI**. Higher grades carry **existing dysplasia**, and **repeat UTIs** → **more scarring** (Fig. 41.7), so **recurrent UTI prompts a VUR work-up**.
- Invasiveness and radiation push towards a **risk-stratified approach**. The **2011 American Academy of Pediatrics (AAP) guideline**: **ultrasound first** after a **first febrile UTI under 2 years**, **VCUG only if ultrasound is abnormal or UTI recurs**.
- VUR found in an **older child or after afebrile UTI** is of questionable significance. Proceed to VCUG for **structural anomaly, renal asymmetry, ureteric dilatation or a bladder anomaly**.
- **Antenatal hydronephrosis is mostly non-refluxing**. Its degree **correlates poorly with VUR grade**, and about a **quarter of refluxing infants have a normal postnatal ultrasound**.

Figure legend, Fig. 41.7 (AUA guideline, 2010) — **More prior UTIs are associated with more baseline renal abnormality**. Sibling- and prenatally detected cohorts have the fewest UTIs and abnormalities.

Impact of Bladder and Bowel Dysfunction

- See VUR as a **primary (innate) component plus treatable secondary elements**. Removing the secondary elements leaves only true primary VUR to treat.
- **IRSC (European arm): 18% had voiding dysfunction**, with **more UTIs and more persistent VUR**. Both **AUA and European Society for Paediatric Urology (ESPU) guidelines** mandate treating BBD.
- **Swedish Reflux Trial**: dysfunction was present in **20% at entry and 34% at 2 years**. Dysfunction at follow-up **correlated negatively with improvement** and with scintigraphic abnormality. **Untreated bladder dysfunction worsens surgical outcome**.
- **AUA (2010; reaffirmed 2017): BBD is one of the most critical modifiable variables**. Meta-analysis: BBD → **more UTI on prophylaxis and after surgery, less resolution at 24 months, lower endoscopic success, but no reduction in open surgical success** (Fig. 41.8).

- **Recognise BBD:** frequency, holding, urgency, incontinence, constipation. Severe BBD shows **curtsying in girls and penile squeezing in boys**. Constipation causes co-contraction of anal and urinary sphincters and **must be eliminated first**.
- **RIVUR plus CUTIE (Careful Urinary Tract Infection Evaluation)** pooled data, off prophylaxis: recurrent UTI in **about half with BBD plus VUR, versus about a fifth with VUR alone and about a third with BBD alone**.

Figure legend, Fig. 41.8 (AUA guideline, 2010) — **BBD is associated with markedly more UTI on continuous antibiotic prophylaxis (CAP) than no BBD.**

Circumcision Status and Impact of Phimosis

Routine circumcision (general population)	Boys with VUR
No overwhelming benefit: number needed to treat (NNT) to prevent one UTI about 111	NNT falls to about 4. The prepuce stays colonised despite CAP. Topical steroid for phimosis, allowing retraction and hygiene, is equivalent to circumcision in preventing UTI

LOWER URINARY TRACT INFECTION AND REFLUX

- Management has shifted from **surgery first** to **medical management first**, with surgery for high-risk, refractory or progressive cases.
- **Reflux does not cause UTI**; it is a **clinical accelerant**, mechanically delivering infected urine to the pelvis.
- **Cystitis** irritates the bladder and lowers the reflux threshold. ***Escherichia coli* endotoxin impairs the antireflux mechanism** and causes **ureteric atony**; in primates, infection made VUR persist.
- **Dilated upper tracts are a reservoir** that recycles colonised urine and makes sterilisation harder.
- **Swedish Reflux Trial:** high-grade VUR on **surveillance alone had a 57% febrile UTI recurrence**; prophylaxis or correction cut this to **about 20%**.

KEY POINTS: URINARY TRACT INFECTION AND REFLUX

- Reflux is not a general cause of UTI.
- Reflux facilitates pyelonephritis.

Concept — Infection, not reflux, is the injury. Reflux is merely the conveyor, which is why sterile urine and good emptying, not the reflux itself, are the targets of treatment.

ASSESSMENT OF THE LOWER URINARY TRACT

Cystographic Imaging

CYSTOGRAPHIC METHODS FOR DETECTING VUR

Method	How	Strength	Weakness
VCUG (gold standard)	Catheter; contrast by gravity ≤ 70 cm H ₂ O; fluoroscopy during filling and voiding	Anatomy and dynamics: grade, volume at onset, intrarenal reflux, diverticula, ureterocele, bladder neck, urethra, postvoid drainage	Catheter; radiation
Direct RNC (gold standard)	Catheter; technetium-99m (^{99m} Tc) pertechnetate; gamma camera	More sensitive for grade II–V; historically about 1% of continuous-fluoroscopy and 10% of pulsed VCUG dose (gap now narrowed); ideal for screening and follow-up	Little anatomy; misses grade I (bladder activity obscures the distal ureter); grading difficult
Indirect RNC	IV tracer; bladder fills from the kidneys; child voids on command	No catheter; sensitivity and specificity 70–90%	Needs a toilet-trained child
Indirect cystography (after excretory urography)	Contrast via the kidneys	No catheter; may suit ruling out VUR	False positives (retained excreted contrast); false negatives for low grades
Contrast-enhanced voiding urosonography	Echo-enhancing agent; colour Doppler	No ionising radiation; comparable to VCUG with second-generation agents; renal ultrasound in the same sitting; detects intrarenal reflux	Grading not standardised; limited availability

- **Postvoid films** are crucial. **Retained pelvic contrast suggests concomitant ureteropelvic junction obstruction (UPJO)**, primary or secondary to massive retrograde filling.
- **Suspect concomitant UVJ obstruction** when VCUG shows: **(1) dilated proximal ureter** (not necessarily kidney); **(2) transition to a distal aperistaltic segment**; **(3) contrast dilution** by trapped urine; **(4) delayed drainage** above that segment. This is a **relative contraindication to endoscopic correction**, which may worsen obstruction.
- **RNC bladder-volume technique:** infusions of about **10% of estimated capacity (Koff formula)** until capacity is reached. The **volume at which VUR first appears predicts resolution**. VCUG and RNC are **complementary**.
- **One cycle can miss VUR: a cyclic study (two or three fill-void cycles) detects 12–20% more.** A limited-image cyclic VCUG reduces dose.

Reflux during filling (passive, low pressure)	Reflux only during voiding (high pressure)
Poor prognostic sign for resolution; common in acquired or neurogenic dysfunction	More favourable; missed if the child cannot void on the table

- **Timing after UTI:** classically **delay VCUG at least 1 week**, because active cystitis risks **iatrogenic pyelonephritis** and infection itself causes dilatation. Current evidence: once the **urine is sterile and the child has improved**, VCUG within about 8 days shows no difference in VUR prevalence.

Cystoscopy and the Positioning of the Instillation of Contrast Cystogram

- **Routine cystoscopy has no role. Orifice appearance and tunnel length correlate poorly with VUR.** Its use is just **before open surgery:** orifice position, duplication, diverticula, inflammation, urethra.

Positional instillation of contrast (PIC) cystography (2003), for **recurrent febrile UTI with a negative VCUG**: general anaesthetic → bladder empty → scope beak close to and facing the orifice → **contrast through the irrigation port from 1 m above the bladder** with simultaneous fluoroscopy → **PIC-VUR = retrograde filling of ureter or pelvis** → empty and repeat on the other side.

- **Treating PIC-detected occult VUR (mostly by endoscopic injection) cut febrile UTI about twentyfold.** PIC grade correlates with **dimercaptosuccinic acid (DMSA)** scarring.
- **But:** it is **non-physiological**, with no age-adjusted pressure, and may create iatrogenic VUR. It needs anaesthesia, so it is **not first line** and only for **rare diagnostic dilemmas** until randomised data exist.

Uroflowmetry and Electromyography

- **European Bladder Dysfunction Study:** little role for invasive urodynamics in UTI or VUR; reserve it for refractory cases.
- **Non-invasive uroflowmetry + postvoid residual (PVR) ± patch pelvic-floor electromyography (EMG):** flow patterns correlate with symptoms and BBD, and **favourable findings predict resolution**. They are worth including in assessment and possibly screening.

Concept — Lower-tract imaging trades **anatomical detail against dose and invasiveness**. VCUG for anatomy, RNC for sensitivity and follow-up, ultrasound contrast to avoid radiation, all made unreliable by a single cycle or an infected bladder.

ASSESSMENT OF THE UPPER URINARY TRACT

Rationale for Serial Assessment of Upper Tracts

- **Pyelonephritis with VUR → scarring, impaired renal growth, renovascular hypertension.** Imaging is selective, guided by reflux status, age, sex, race, family history and bladder–bowel function.
- The core goal is to **distinguish VUR-related damage from congenital dysplasia, medical renal disease or hydronephrosis**, which is best done by **serial** imaging.

Renal Sonography

- **RBUS is the initial mainstay:** non-ionising, ideal for serial use. It measures **renal volume against growth curves**, shows the impact of intercurrent febrile UTI, and guides the need for scintigraphy or correction. **Renal length correlates well with scintigraphic hypoplasia.**
- **Loss of corticomedullary differentiation or increased echogenicity** → functional abnormality. In a neonate with high-grade VUR and **no UTI**, plus a small kidney, this indicates **congenital dysplasia, not acquired scar**.
- **Limits:** ultrasound cannot diagnose VUR, and a **normal scan does not exclude it**. It is **insensitive to small cortical defects** (scintigraphy is superior) but adequate for surveillance, large defects and size asymmetry.

Renal Scintigraphy

- **Gold standard for functional parenchymal imaging:** ^{99m}Tc -DMSA, taken up only by **functioning proximal tubules**, in proportion to glomerular filtration.
- Uses: **reconstruct versus remove** when function is minimal; **conjugate views** for differential function with severe hydronephrosis.
- **Acute pyelonephritis → photopenic areas** (oedema impairs tubular uptake). Many resolve with prompt treatment; **persistence = scar**.

- **Seminal DMSA study: 98% sensitivity, 92% specificity** for scar. Meta-analysis: **VUR about triples the risk of cortical abnormality after infection**, and **about 30% of children with pyelonephritis have VUR**.
- **Single-photon emission computed tomography (SPECT)-DMSA**: three-dimensional, slightly more sensitive, better reader agreement. RIVUR showed **large inter-institutional variability**, with a sizeable share of scans rejected for quality.
- **DMSA shows congenital maldevelopment** in high-grade VUR without any UTI, so the **congenital-versus-acquired distinction is critical**.

Figure legend, Fig. 41.10 — Pinhole DMSA: a normal left kidney and a right kidney with **multiple cortical defects**.

CORTICAL DEFECTS

Congenital Defects Versus Acquired Scar

- A scar is a **fibrous, contracted region destroyed, usually, by infection**: photopenic on DMSA, and hyperechoic and shrunken on ultrasound when large.
- Reflux-associated **dysplasia looks identical**, and the term “scar” has become **contaminated** by including non-preventable congenital dysplasia.

	Reflux-associated renal dysplasia	Acquired renal scar
Cause	Abnormal bud–blastema interaction (Mackie–Stephens)	Pyelonephritis: infected urine plus VUR
Typical patient	Male infant, grade IV–V, antenatal hydronephrosis, no UTI	Female, recurrent febrile UTI, BBD
Imaging	Small kidney, globally reduced function, global or focal poor uptake	Focal photopenic defects after documented infection
Preventable?	No	Yes: keep urine sterile
Also seen in	Duplex moieties, prune-belly syndrome, PUVs	—

Reflux-Associated Renal Dysplasia

TABLE 41.4 CONGENITAL RENAL SCARRING

Grade of VUR	Normal kidney	Slight damage	Severe damage
I–III	All	None	None
IV	About half	About a third	Minority
V	Minority	About a third	Nearly half

Modified from Marra G, Barbieri G, Dell'Agnola CA, et al., J Pediatr 1994 (prenatally detected VUR in male infants).

- **What it tests**: in prenatally detected boys, **before any infection, renal damage is confined to grades IV–V and rises steeply with grade**. The trap: such damage is **congenital, not “scarring” from infection**, and preventing UTI will not reverse it. **Boys are spared infective scarring** by their lower rate of recurrent UTI.

Figure legend, Fig. 41.12 — Massive VUR with antenatal hydronephrosis. Ultrasound shows **poor internal differentiation and no function on scan**.

Acquired Renal Scars

- **Nephropathy from primary VUR requires bacterial colonisation** with normal voiding. **Sterile VUR damages only with secondary high pressures (PUVs)** or rare megacystis states.

- **Ransley and Risdon (pig model)**: reflux created by unroofing the tunnel caused **nephropathy only if the urine was infected**. Clinical series confirm more scarring with multiple UTIs.
- **High pressure modulates sterile reflux**: higher voiding pressures raise scarring risk. **Hodson**: VUR plus outflow obstruction caused **atrophic pyelonephritis without infection**, worse with bacteria. Hence **normalising bladder and bowel function is essential** while awaiting resolution.

Pathophysiology of Acquired Scarring

- **VUR is an accelerant** after bladder colonisation: **more pyelonephritis with high grade**. After correction, **almost no new scars despite asymptomatic bacteriuria**, confirmed by meta-analysis.

Age

- **Scarring propensity is inversely related to age**. **Highest risk in the first year; under 4 years is more prone after a single UTI**, though scarring still occurs beyond 5. In older children, scarring usually reflects **late diagnosis, inadequate treatment or social factors**.
- **“Big bang” theory (Ransley and Risdon)**: most scarring follows the **first pyelonephritis**, with little further change without new infections. It is challenged by **new scars in landmark trials, dysplasia mistaken for scar, defects changing with growth, and weak evidence for greater infant vulnerability**.

KEY POINTS: CORTICAL DEFECTS

- **Sterile VUR is unlikely to scar.**
- **The youngest are at greatest risk** of post-pyelonephritic scarring.
- Most parenchymal abnormality is **dysplasia or pyelonephritis**, and these are **hard to tell apart**.

Host Susceptibility and Response

- Bladder colonisation depends on **periurethral flora** (vaginal and preputial), **hygiene and bowel habit** (the gut is the reservoir) and **host adherence receptors**.
- **Under 6 months, most UTIs in either sex occur in uncircumcised boys**, reflecting dense *E. coli* colonisation of the infant prepuce.
- **Urinary dwell time is the greatest tangible factor**. *E. coli* **doubles in minutes**, so a bladder **not emptied for hours** incubates a clinical inoculum. **Regular emptying is fundamental**.

Papillary Anatomy

Compound papillae (concave)	Simple papillae (convex)
Ducts open at right angles, so there is no valve	Ducts open obliquely , creating a valvular effect
Concentrated at the poles → intrarenal reflux and polar scars	Mostly mid-kidney
Reflux into ducts at lower pressure	Higher pressure needed

- **Intrarenal VUR occurs at only 2 mm Hg in the neonate**; by 1 year, an **order of magnitude more** is needed, so it is rare in older children (Fig. 41.14).

Bacterial Virulence

- **Fimbrial adhesion** to uroepithelial receptors is the chief virulence trait. Yet scars **with VUR** often arise from organisms with **fewer than three virulence factors: reflux lets less virulent bacteria scar**. Urinary metabolic and microbiome factors are emerging.

Hypertension

- **After essential hypertension, reflux nephropathy may be the commonest cause of hypertension in children**, via the **renin-angiotensin system** and $\text{Na}^+/\text{K}^+\text{-ATPase}$. Severity of nephropathy correlates with progression on ambulatory monitoring.
- **34–38% of children with renal scarring** become hypertensive in long-term adult follow-up. VUR first diagnosed in adulthood is strongly associated with hypertension.
- It is unclear whether **acquired scar, congenital dysplasia or both** drive it. If dysplasia already exists, preventing infection may not prevent hypertension.
- **Correcting VUR does not lower blood pressure**, because the mechanism is **parenchymal microvascular**. **Segmental resection guided by selective renal vein renin**, or **nephrectomy of a small dysplastic or shrunken kidney**, can cure it in selected patients.

Renal Growth

- **VUR correction poorly predicts catch-up growth**. **Contralateral compensatory hypertrophy** exaggerates the apparent ipsilateral deficit. The **worst growth occurs with VUR plus UTI in a solitary kidney**.
- **No superiority of medical over surgical therapy** for growth or scarring. **IRSC at 10 years: no difference in somatic growth**. Surgery may help when **BUTI shows prophylaxis has failed**.

Renal Failure and Somatic Growth

- **Renal failure from primary VUR is now rare**, thanks to Smellie's paradigm (1970s–1980s): prophylaxis and infection control. **United States Renal Data System (USRDS)**: reflux nephropathy as a cause of renal replacement therapy (RRT) fell from **1.9 to 0.6 per million (1996–2014)**.
- Yet reflux nephropathy is the **second commonest primary diagnosis in non-Black paediatric transplant recipients**, and in Australia and New Zealand the **second cause of paediatric RRT after glomerulonephritis**. Some shift may be **reclassification as congenital dysplasia**.
- **Medical renal disease with scarring**: hyperfiltration, **concentrating defect**, proteinuria, microalbuminuria, renal tubular acidosis, raised fractional sodium and magnesium excretion. **Concentrating defects occur even with sterile VUR**, are proportional to grade and **improve after cessation**, suggesting a **functionally obstructive** element.
- **Somatic growth is a global marker**: children with bilateral VUR and damage fall below centiles, and suppressing pyelonephritis allows **catch-up**. Prophylaxis was associated with a higher body mass index (BMI) over 1 year and a non-significant fall in linear growth under 1 year.

Concept — Every downstream harm of reflux, whether scar, hypertension, growth or renal failure, splits into a **preventable infective component** and a **fixed congenital component**. Treatment can only ever act on the first.

Controversies in Usage of Invasive Investigations

- **Catheterisation distress** and the arrival of **low-morbidity endoscopic correction** have challenged convention. Every assumption is disputed: scarring incidence, the VUR-renal failure link, the age of peak susceptibility, and whether resolution is even necessary.
- **Distress mitigation: sedation, topical urethral anaesthetic, hypnosis**. Each test must be justified by how it changes the decision tree.

GUIDELINE APPROACHES TO IMAGING AFTER UTI

	NICE (UK)	AAP (2011)	Top-down approach (TDA)
Population	Infants and children with UTI	Febrile UTI, 2–24 months	Febrile UTI
First test	Ultrasound	Ultrasound	DMSA soon after febrile UTI
VCUG when	<6 months: atypical or recurrent UTI with ultrasound abnormality. 6 months–3 years: atypical or recurrent (dilatation, poor stream, non-<i>E. coli</i> , family history); none if response within 48 h. >3 years: no routine VCUG after a first UTI	Not after a first febrile UTI unless ultrasound shows hydronephrosis, scarring or dysplasia, obstruction, or the case is atypical; after a second febrile UTI	Only if DMSA is abnormal , or with recurrent UTI. Follow-up DMSA at 4–6 months if pyelonephritis is uncertain
Other changes	No early DMSA	Culture threshold lowered to 50,000 colony-forming units (CFU)/mL plus abnormal urinalysis; oral = parenteral antibiotics	Acute-phase DMSA adds little over delayed-phase
Weakness	Misses much VUR, including grades IV–V, and many scars; no outcome validation	The AAP Section on Urology objected: underpowered trials, mixed collection, ignores BBD , assumes a true first UTI. VCUG use fell sharply	DMSA is sensitive but not specific: misses about a third of VUR, mostly low grade; normal DMSA → <1% chance of high-grade VUR ; about 70% of febrile UTIs have abnormal DMSA (over-labelled high risk); access, cost, radiation

- **What the table tests:** all three reduce VCUG use by choosing a **gatekeeper test**. Ultrasound (NICE, AAP) is **specific but insensitive** for VUR and scar. DMSA (TDA) is **sensitive for high-grade VUR but non-specific**. A **normal DMSA effectively excludes high-grade VUR**; an **abnormal one proves little**. The trap: guideline compliance is not the same as safety for the individual child.

Concept — The guideline debate is a debate about **which first test should gate the catheter**. Each choice trades missed reflux against avoided distress, and none replaces individual risk assessment.

ASSOCIATED ANOMALIES AND CONDITIONS**Ureteropelvic Junction Obstruction**

- **VUR in UPJO:** up to about 5–9% in contemporary series. **UPJO in VUR: 0.75–3.6%**, and **high-grade VUR is five times more likely to coexist with UPJO**. Half of high-grade sonographic hydronephrosis with high-grade VUR showed an **obstructive diuretic renogram**. A shared **ureteric bud anomaly** is proposed.
- Coincidental VUR with UPJO is usually **low grade and resolves**. A **minimally dilated ureter with a huge pelvis** means the VUR grade is **overestimated**.
- **Four VCUG signs of UPJO with VUR:** (1) dilated ureter with little pelvic filling; (2) dilute, faint pelvic contrast; (3) ureter–pelvis dilatation discrepancy; (4) pelvis that fails to drain postvoid.
- **Correct the obstruction first** (confirmed by scintigraphy with catheter drainage). Sterile VUR can be observed, but obstruction harms even without infection. Repairing VUR first risks **worse dilatation from lower-ureter oedema** and **infection of an obstructed pelvis**.
- **Simultaneous open repair of both** threatens ureteric blood supply. **Endoscopic injection at pyeloplasty is reasonable**, despite lower success.

Figure legend, Fig. 41.15 — VUR fills the ureter to the UPJ with **minimal pelvic filling**, a sign of UPJ obstruction. **Kinking of the UPJ can appear only during voiding**.

Ureteral Duplication

- **VUR is the commonest abnormality in complete duplication, usually into the lower pole.** The **Weigert-Meyer rule**: the lower-moiety orifice is **more lateral and proximal**, with a **shorter intramural tunnel**.
- Correction was traditionally favoured (slower resolution), but **duplication alone does not raise the risk of grade progression, BUTI or scarring**, so **conservative management is reasonable**. Low grades resolve at high rates. **High-grade lower-pole VUR in girls** is more prone to BUTI and scarring and may warrant treatment.

Figure legend, Fig. 41.16 — Reflux into both ureters of a duplex is less common than lower-pole VUR alone.

Bladder Diverticula

Paraureteral diverticulum (ureter enters bladder beside it)	Ureter enters the diverticulum
Common with VUR and may raise UTI risk, but resolution is the same as primary VUR (grade is the only predictor). Manage as for the VUR itself	No muscle backing → VUR will not resolve . Decide on repair by the impact of the VUR and the diverticulum's size and shape

- Hutch: paediatric diverticula are **congenital**, in smooth-walled normal bladders.

Renal Anomalies

- **Multicystic dysplastic kidney (MCDK): contralateral VUR in about a quarter**, half low grade. It **resolves spontaneously** (mean about 4 years) regardless of grade, and surgery is rarely needed. Contralateral VUR **blunts compensatory hypertrophy**.
- **Renal agenesis: contralateral pathology 32–46%. VUR is the commonest (24–28%)**, then UVJ and UPJ obstruction. Meta-analysis suggests about 17% (close to background), with a high share dilating, half of which resolve. **Knowing it rarely changes management**.

Megacystis-Megaureter Association

- **Massive bilateral VUR**: urine voided partly up the ureters returns to the bladder after voiding → **a large thin-walled bladder and massive hydroureter** (Williams; Burbige). It is **detectable in utero** and **commoner in males**.
- It is **non-obstructive**, like cardiac dilatation from valvular regurgitation. **VCUG shows a normal posterior urethra**, distinguishing it from **PUVs or prune-belly syndrome**.
- Large residuals → recurrent UTI. **Temporise with vesicostomy or cutaneous loop ureterostomy → reimplant** (UVJ failure is the primary lesion) → postoperative **bladder rehabilitation**, after which volume and contractility normalise.
- **Contraindicated: procedures on an already patent outlet**, which risk infection and miss the cause.

Obstructive Refluxing Megaureter

- It **mimics high-grade dilating VUR**, and contrast may not ascend, so it is **under-graded**. **Key sign: contrast in the ureter much more dilute than in the bladder**.

- **Endoscopic injection may worsen hydroureteronephrosis and obstruct.** CAP may reduce UTI, but data are sparse. Options include endoureterotomy plus injection (limited data) or **reimplantation**, refluxing or non-refluxing, $\pm$ tapering.

Other Anomalies

- **Over 40 syndromes** include VUR (Online Mendelian Inheritance in Man), notably **VACTERL** (vertebral, anal, cardiac, tracheo-oesophageal, renal, limb), **CHARGE** (coloboma, heart disease, choanal atresia, retarded development, genital hypoplasia, ear anomalies) and **imperforate anus**. **VCUG is the initial study** when VUR is anticipated.

Concept — In each association, **the coexisting lesion sets the priority**. Obstruction is corrected before reflux, a duplex or diverticulum is managed as for the reflux unless the anatomy precludes resolution, and a non-obstructive megacystis is cured by fixing the UVJ, not the outlet.

FOLLOW-UP AND TRANSITIONING INTO ADULTHOOD

- **Childhood VUR carries a persistently higher adult UTI risk**, even after resolution or correction. In women, **BBD persisting into adulthood** predicts febrile and afebrile UTI regardless of VUR status.
- **Dextranomer/hyaluronic acid copolymer (Dx/HA) mounds calcify in over half** and **mimic stones**. After **cross-trigonal or advancement reimplants**, retrograde catheterisation and ureteroscopy are harder. **Knowing the surgical history avoids misadventure**.

Pregnancy and Reflux

- **Pregnancy reduces bladder tone** (oedema, hyperaemia) and causes **physiological upper-tract dilatation**. Slow drainage favours bacteriuria and pyelonephritis.
- **VUR in late pregnancy $\rightarrow$ far more pyelonephritis** (about a third versus under 5% without VUR).
- **Prior UTI $\rightarrow$ bacteriuria; scarring plus persistent VUR $\rightarrow$ pyelonephritis**.
- **Renal scars $\rightarrow$ about threefold hypertension, about eightfold pre-eclampsia**, and more obstetric intervention. **Bilateral scars carry more pre-eclampsia than unilateral**.
- **Normal blood pressure and renal function $\rightarrow$ uneventful pregnancy; impaired function $\rightarrow$ fetal loss and accelerated maternal renal disease**.
- **Reimplanted in childhood $\rightarrow$ still at UTI risk in pregnancy but not at excess miscarriage risk**. Mother and fetus are at significant risk only with **scarring or hypertension**.
- **Meta-analysis: renal scarring, not the presence of VUR, drives pregnancy morbidity**. Long-term follow-up links scarring to UTI, pregnancy hypertension and **low birth weight**. **About 43% of offspring** who had a VCUG had VUR.
- **Persistent VUR without scars: risk is poorly defined**. Some **correct VUR persisting beyond puberty in girls**, while others stop prophylaxis and monitor. **Long-term follow-up and transition programmes are essential**.

KEY POINTS: PREGNANCY AND REFLUX

- **Pregnancy physiology** increases VUR-associated pyelonephritis risk.
- **Correction of VUR may be considered before pregnancy**, absent definitive evidence otherwise.

Concept — In adult life, **the scar, not the reflux, carries the risk**. That is why correction does not erase the pregnancy and UTI hazard and why women with childhood VUR need lifelong follow-up.

NATURAL HISTORY AND MANAGEMENT

Spontaneous Resolution

- Resolution is **roughly inversely proportional to initial grade**. Later in life, it depends on **initial grade and age at presentation**.
- Mechanism: **remodelling of the UVJ** (elongation of the intravesical ureter) plus **stabilising bladder dynamics**. **Persistent BBD explains failure to resolve**.

Resolution by Grade

SPONTANEOUS RESOLUTION BY GRADE

Grade	Expected resolution
I-II	Most resolve ; grade II 63–85% across studies
III	About 50% ; about 90% of resolutions occur within 4 years
IV-V, bilateral III	Few (IRSC: uniformly low)

- Large retrospective nomogram series show the same gradient. Grade assignment varies by at least one grade, so **fine distinctions above grade III are of little clinical relevance**.

Resolution by Age

- **Age at presentation matters more than grade**. VUR is commonest and most likely to resolve in **neonates and young children**.
- **AUA 1997 guideline** estimates by age and grade (Fig. 41.18), confirmed in the 2010 revision and 2017 validation, refocused on risk factors beyond age.
- **High grades can resolve when diagnosed early**: infants with grade IV–V resolved at rates close to low grades in the first year. One prospective infant cohort with grade III–IV resolved fastest in year 1, then slowed to about **9% per year**. Conservative management has a role even in high-grade infant VUR.
- **Traditional observation period: 5 years**. Reported resolution rates vary widely between series, reflecting unmeasured variables. **Each child's risk is dynamic**.

Figure legend, Fig. 41.18 (Elder et al., AUA Pediatric VUR Guidelines Panel, 1997) — Persistence over 5 years is lowest for grade I, then II, higher for unilateral grade IV and highest for bilateral grade IV. For grade III, older age and bilaterality increase persistence: unilateral age 1–2 resolves most, bilateral age 6–10 least.

LANDMARK STUDIES

- Studies split by grade (**low I–III, high IV–V**) have not produced a unifying role for CAP. **Edwards (1977)** first showed that **low-dose prophylaxis prevented infection and scarring while VUR resolved**.

RANDOMISED AND PROSPECTIVE TRIALS OF PROPHYLAXIS AND CORRECTION

Trial	Population	Comparison	Key result	Verdict
IRSC (North American and European)	<9 years, high-grade VUR	CAP vs open surgery	UTI 38% in both arms ; surgery reduced pyelonephritis more (some transient postoperative obstruction); new scars equal . At 10 years: more febrile UTI medically, but equal growth, imaging and renal function ; most scars in the first 5 years; VUR kept declining	Surgery ≈ medical for scarring
Birmingham Reflux Study	High-grade	Medical vs surgical, 5 years	New scars equal, all within 2 years ; over half still refluxing at 5 years	Equal; supports “big bang”
Swedish Reflux Trial	1–2 years, grade III–IV	CAP vs endoscopic injection vs surveillance, 2 years	Girls : febrile UTI recurrence 19% CAP, 23% endoscopic, 57% surveillance ; new scars mostly on surveillance, some after endoscopy, none on CAP . Boys : no difference . BBD → less resolution	Treat girls with dilating VUR (CAP or injection)
RIVUR	607 children, grade I–IV , after first or second febrile or symptomatic UTI; >90% girls, mostly I–III	Trimethoprim-sulfamethoxazole (TMP-SMX) vs placebo , double-blind	Recurrent UTI halved: 23.6% → 12.9% , greatest with febrile index UTI or BBD . Renal scarring unchanged (11.9% vs 10.2%). Placebo still achieved 90% treatment success	CAP prevents UTI, not scars ; benefit concentrated in high risk
PRIVENT	576 children with UTI, mixed VUR status	CAP vs placebo	Absolute UTI reduction 6% (19% → 13%) across subgroups	Modest benefit
Southwest Pediatric Nephrology Group	Grades I–III	Medical only	Most resolved; a third had BUTI; new scars more frequent in grade III	Medical management is not risk-free
Randomised trial, pyelonephritis ± VUR	Low-risk, at most grade III	CAP vs observation	No difference ; recurrence equal with and without VUR	No benefit at low risk
French multicentre trial	Grades I–III, 1–3 years	CAP vs none	No significant difference ; possible benefit in boys with grade III	No benefit at low grade
Small randomised trial	Grades II–IV after first UTI	CAP vs none	No difference in UTI or scarring	Underpowered

- **Reconciling the trials:** RIVUR was **diluted by low-risk patients** (only about 8% grade IV). Reanalysis split into **high and low risk: NNT 5 in high-risk versus 18 in low-risk**, with **no significant benefit at low risk**. Machine-learning models now allocate CAP using RIVUR data.
- **Why the trials disagree: population risk.** Trials enrolling high-grade, female, BBD or febrile cohorts show benefit; low-grade and mixed cohorts do not.

- **Early trial flaws:** small numbers, heterogeneous groups, high dropout, **bag urine samples**, no placebo, routine screening cultures, and no distinction between febrile and afebrile UTI. RIVUR itself had poor adherence and withdrawals.

Concept — *The landmark trials do not contradict each other; they **sampled different risk**. Prophylaxis reduces infection in proportion to baseline risk, and **no trial shows that either prophylaxis or surgery prevents scarring better than the other**.*

CURRENT APPROACH TO MEDICAL MANAGEMENT

Risk Stratification

- **One principle supersedes all: preserve renal function and prevent UTI and scarring.** After the 2011 AAP shift and fewer cystograms, practice is **risk-based**. The tension comes from open correction succeeding in **over 95%**, and robotic correction being **likely less morbid**.

Eight tenets: (1) **most VUR resolves**, helped by correcting BBD → (2) **higher grades resolve less**, especially in older children after UTI → (3) **sterile VUR rarely damages** → (4) **preventing UTI matters more than resolving VUR** → (5) **prophylaxis is safe and beneficial, especially at high risk** → (6) **medical management suits most VUR** → (7) **multivariable risk stratification** predicts resolution, UTI and nephropathy → (8) **long-term follow-up**, because VUR is dynamic.

- **Evidence summary:**
 - **Little CAP benefit at grade ≤II**, especially without BBD; **benefit at grade ≥III, at least in girls**.
 - **About 15% have a recurrent febrile UTI within 2 years**, and **about 15% of those scar**.
 - **BBD → recurrent UTI in about 45% versus 15%** without, on or off antibiotics.
 - **Higher grade → more pyelonephritis and new damage**. The effect of age is uncertain.
 - **Uninterrupted follow-up**, especially with scars: hypertension, medical renal disease, and **women through childbearing years**. Transition programmes support this.
- **Classic approach:** daily low-dose CAP while awaiting resolution, whatever the grade. **Extensive scarring, high grade, globally reduced function or dysplasia** lower tolerance for another infection and favour correction.
- **Stopping CAP** after toilet training or once BBD resolves: **about 8–10% then develop febrile UTI**, which mandates reassessment and possible correction. Select low-risk children prospectively. Most children with asymptomatic VUR need little follow-up beyond **lifelong bladder and bowel habits and prompt review of any UTI**.
- **Unresolved controversy:** asymptomatic bilateral high-grade VUR with a normal ultrasound in an older boy. Girls were traditionally corrected after toilet training for **pregnancy reasons**, an approach not yet proven.

Current Indications for Continuous Antibiotic Prophylaxis

- **“Water-hammer” sterile renal damage has never been reliably demonstrated.** What is managed is infection risk, not VUR.
- **Daily low-dose oral CAP, preferably at night**, which gives the longest bladder concentration during the longest retention, when infection is likeliest.

PROPHYLACTIC AGENTS: USE AND AGE

Agent	Age	Advantage
Trimethoprim	<2 months (first choice) and older	Avoids sulfonamide in the neonate
Amoxicillin	<2 months	Safe in the neonate
TMP-SMX (co-trimoxazole)	>2 months	Common, well tolerated; agent used in RIVUR
Nitrofurantoin	>2 months	Less resistance in faecal flora (the pathogen reservoir)
Probiotics (<i>Lactobacillus</i>)	Infants after pyelonephritis	Displace uropathogens. Alone: no proven benefit on meta-analysis; possible adjunct to antibiotics

PROPHYLACTIC AGENTS: SIGNATURE TOXICITIES

Agent	Signature toxicity	Implication
Sulfamethoxazole (in TMP-SMX)	Displaces bilirubin → neonatal jaundice (hepatic immaturity)	Avoid under 2 months
TMP-SMX	Transient leukopenia, rare and reversible	No routine blood-count monitoring
Nitrofurantoin	Interstitial lung disease (pulmonary fibrosis, pneumonitis); hepatic; poor taste, gastrointestinal upset	Serious toxicity is rare and mostly with full-dose , not low-dose, therapy

- Check **local susceptibility**; agents and doses vary regionally.
- **Breakthrough febrile UTI or pyelonephritis on CAP, despite good bladder and bowel habits → stop medical management and correct the VUR.** Confirm proper urine collection first.
- **Smellie's interpretation of BUTI:** an organism **sensitive** to the prophylactic → **non-adherence or too low a dose**; an organism **resistant** → **residual volume too high too often, or the dose too high**.
- **Hidas risk model** (presentation with UTI, high grade, female, BBD), validated prospectively: **2-year BUTI on CAP 8.6% low, 26% intermediate, 62.5% high risk**. Other tools (VURI, UDR-based, machine learning) incorporate grade, laterality, age, BBD, number of infections, presentation, **bladder volume at onset**, UDR and scarring.
- Several trials question CAP even at high grade. Stopping CAP, or failing endoscopy while staying off CAP, does **not** raise UTI rates in selected children.
- **Core message:** identify the child's **current risk**, including social and behavioural factors. Teach **perineal hygiene, timed emptying, and diet and bowel measures**, check adherence and access, **start CAP for high risk**, and **move children between risk strata at each visit**.

Individualizing Patient Care

- **Minimum initial work-up:** an accurate history of the **number and severity of clinically significant UTIs** plus **high-quality renal and bladder ultrasound**; then decide on VCUG.
- **Multiple documented febrile UTIs, admissions or upper-tract findings → VCUG imperative.** Afebrile “infections” with doubtful urine studies and a normal ultrasound → a conservative approach with little risk.
- **Risk stratification** (Fig. 41.19): in validation, about two-thirds of children with VUR were **low risk** (BUTI about 9%), about a quarter **intermediate** (about 27%) and only a few **high risk** (about 63%). This mirrors the low-risk predominance of all-comer cohorts such as RIVUR.

FIG. 41.19 RISK FACTORS AND RISK GROUPS FOR VUR

Risk factor	Low risk	High risk
Sex	Male	Female
Age	>5 years	<1 year
Race	Black	White
Laterality	Unilateral	Bilateral
Grade	I–II	III–V
BBD	No	Yes
Scars	No	Yes
Presentation	Prenatal hydronephrosis	Multiple febrile UTIs

Modified from Hidas G, Nam A, Soltani T, et al., Arab J Urol 2013.

- The aim is to **prevent infection while allowing time for spontaneous resolution**, with management **specific to each child and adaptable as risk changes**.

KEY POINTS: NATURAL HISTORY AND MANAGEMENT

- **Reflux tends to resolve** given time; likelihood is **inversely proportional to grade** but modified by other factors.
- **Risk stratification and individualised care** are central.
- **Urine sterility**, by prophylaxis and strict bladder and bowel management, is the **cornerstone of watchful waiting**.
- **CAP reduces UTI but not necessarily scarring or chronic kidney disease (CKD)**.
- **CAP benefits most:** higher grade, bladder dysfunction or BBD, febrile presentation, higher UDR, high validated risk scores.

Concept — Modern VUR management treats **infection risk, not reflux**. Grade is one input among many, prophylaxis is aimed at the high-risk minority, and **the bladder and bowel are treated first because they move a child between risk strata**.